BILINEAR RANK-2 FACTORIZATION OF DIRICHLET ENERGY KERNELS, REVERSE-TRIANGLE AND SIEVE-HORIZON OBSTRUCTIONS, PREISSMANN–MONTGOMERY–VAUGHAN L^2 BOUNDS, AND FAILURE OF POINTWISE GALLAGHER TRANSFER

THIJS MISSIAEN

Abstract. In the symmetric Riemann–Siegel model of Riemann's Ξ-function at $z = t - \mathrm{i}d$ ($d = \sigma - \frac{1}{2} > 0$), we establish an exact rank-2 factorization of the integrated transverse Wronskian cross-term kernel $w_{mn}^{\mathrm{int}}(d,t) = \frac{2}{d^2}[\cosh(d\omega_m(t))\cosh(d\omega_n(t)) - 1]$, reducing $\mathcal{O}(N(t)^2)$ off-diagonal double sums to $\mathcal{O}(N(t))$ prefix sums. Combining w_{mn}^{int} with the critical-line boundary term cancels the -1 offset pair by pair and yields the Euclidean representation $\frac{1}{4}|S_N(d,t)|^2 = \|\psi_1(d,t) + \Psi_{\geq 2}(d,t)\|^2$. Under the polynomial sieve $P_r(x) = 1 - (1-x)^r$, we distinguish the squarefree-divisor sieve $\widetilde{\Lambda}_r(n)$ from Bombieri–Selberg's $\Lambda_r^{\mathrm{BS}}(n)$ and show that $b_n^{(r)}(X) = \widetilde{\Lambda}_r(n)/(\ln X)^r$ fails for $n > X$. Moreover, because $\|\psi_1(d,t)\|_{\min} = \sinh(dL/2) \ll \Sigma_{\cosh}^{(r)}(d,t)$ on $0 < d < \frac{1}{2}$, phase-blind reverse-triangle bounds collapse to zero. Finally, using Preissmann's Hilbert constant $c_{\mathrm{MV}} = \pi\sqrt{1 + \frac{2}{3}\sqrt{\frac{6}{5}}}$, we prove that $1 + 2d > 1$ yields Bohr–Landau–Carlson mean-square suppression on windows $\Delta T \gg N(T)^{1-2d}$, whereas Gallagher's Sobolev term diverges as $N(T)^{1-2d}$, precluding pointwise positivity.

1. Introduction and Canonical Wronskian Framework

Let $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with transverse distance $d = \sigma - \frac{1}{2} \geq 0$ and complex spectral coordinate $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. Throughout this series [11, 12, 13], Riemann's completed entire function $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its symmetric counterpart $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$ [17, 20] are governed by the canonical Fourier–Jacobi normalization

$$\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u = 4\int_{-\infty}^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u, \tag{1.1}$$

where $\Phi(u) := \sum_{m=1}^\infty (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u})\exp(-\pi m^2 \mathrm{e}^{4u})$ is the even Jacobi theta kernel [9, 15]. Writing $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ with R, I real-valued, the canonical transverse Wronskian and its parity-reduced counterpart are

$$W(d,t) := \frac{1}{2}\partial_d\big|\Xi(t - \mathrm{i}d)\big|^2 = RI_t - IR_t = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(d,t) := \frac{W(d,t)}{d}, \tag{1.2}$$

with autocorrelation kernel $I(y,t) := \int_{-\infty}^\infty \Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$ [22]. By the fundamental theorem of calculus on $[0,d]$,

$$\frac{\big|\Xi(t - \mathrm{i}d)\big|^2}{d^2} = \frac{\Xi(t)^2}{d^2} + \frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u. \tag{1.3}$$

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ORCID: 0009-0007-3678-0810.

The purpose of this paper is twofold: first, to isolate the genuine algebraic identities that simplify Dirichlet energy kernels attached to the symmetric Riemann–Siegel main sum of length $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$; and second, to prove the exact arithmetic and analytic obstructions that prevent any phase-blind or Sobolev-transferred version of these identities from yielding pointwise zero-free regions in the interior strip $0 < d < \frac{1}{2}$.

Specifically, we establish four rigorous results, whose architecture is summarized below.

1.1. **Proof overview.**

(1) *Sieve function $\widetilde{\Lambda}_r(n)$ versus Bombieri–Selberg $\Lambda_r^{\mathrm{BS}}(n)$, and breakdown for $n > X$ (Section 2).* Under the polynomial mollifier $M_X^{(r)}(s) = \sum_{k\le X}\mu(k)P_r(\frac{\ln(X/k)}{\ln X})k^{-s}$ with $P_r(x) = 1-(1-x)^r$, the product $\zeta(s)M_X^{(r)}(s) = \sum_{n=1}^{\infty} b_n^{(r)}(X)n^{-s}$ $(\sigma > 1)$ satisfies $b_n^{(r)}(X) = \widetilde{\Lambda}_r(n)/(\ln X)^r$ on $2 \le n \le X$, where $\widetilde{\Lambda}_r(n) := -\sum_{k|n}\mu(k)(\ln k)^r$ generates $(-1)^{r+1}\zeta(s)(\zeta^{-1})^{(r)}(s)$. We prove all structural laws of $\widetilde{\Lambda}_r(n)$ (vanishing for $\omega(n) > r$, parity alternation, sharp radical bound $|\widetilde{\Lambda}_r(n)| \le (\ln \mathrm{rad}(n))^r$, and unsigned Stirling asymptotics), distinguish it from Bombieri–Selberg's generalized von Mangoldt function $\Lambda_r^{\mathrm{BS}}(n) := \sum_{d|n}\mu(d)(\ln(n/d))^r \ge 0$ [2, 8], and prove that $b_n^{(r)}(X) = \widetilde{\Lambda}_r(n)/(\ln X)^r$ fails at $\mathcal{O}(1)$ scale for $n > X$ (in particular $b_p^{(r)}(X) = 1$ for primes $p > X$). Consequently, substituting $\widetilde{\Lambda}_r(n)/(\ln X)^r$ on $2 \le n \le N(t)$ requires $X \ge N(t)$ (where the approximate functional equation incurs a convolution tail up to $N(t)X$, as proved in Paper VI [12]) and cannot self-consistently define an $X < N(t)$ threshold.

(2) *Exact bilinear rank-2 factorization of integrated Wronskian cross-kernels (Section 3).* Integrating the hyperbolic cross-term kernel of a symmetric harmonic model over $u \in [0, d]$ yields the exact rank-2 factorization $w_{mn}^{\mathrm{int}}(d,t) = \frac{2}{d^2}[\cosh(d\omega_m(t))\cosh(d\omega_n(t)) - 1]$, reducing $\mathcal{O}(N^2)$ off-diagonal double sums to $\mathcal{O}(N)$ linear sums without approximation.

(3) *Euclidean pure-square identity and the* $\sinh(dL/2)$ *reverse-triangle obstruction (Section 4).* Adding the critical-line boundary term cancels the -1 offset in $w_{mn}^{\mathrm{int}}(d,t)$ pair by pair and yields $\frac{1}{4}|S_N(d,t)|^2 = \|\psi_1(d,t) + \Psi_{\ge 2}(d,t)\|^2$ in $\mathbb{R}^2$. However, $\|\psi_1(d,t)\|^2 = \sinh^2(dL/2) + \cos^2\phi_1(t)$ attains its minimum $\|\psi_1(d,t)\|_{\min} = \sinh(dL/2)$ (not $\cosh(dL/2)$) whenever $\cos\phi_1(t) = 0$. Because $\Sigma_{\cosh}^{(r)}(d,t)/\sinh(dL/2) \asymp N(t)^{1/2-d}(\ln N(t))^{r-1}(\ln X)^{-r} \to +\infty$ for every fixed $d \in (0, \frac{1}{2})$ at $X = N(t)$, the true reverse-triangle lower bound $\max(0, \|\psi_1\|_{\min} - \|\Psi_{\ge 2}\|)$ collapses identically to zero for all large t.

(4) *Preissmann–Montgomery–Vaughan L^2 window bounds versus pointwise Gallagher explosion (Section 5).* Using Preissmann's true sharp constant $c_{\mathrm{MV}} = \pi\sqrt{1 + \frac{2}{3}\sqrt{\frac{6}{5}}} \approx 4.1324743621$ [16] in the Montgomery–Vaughan Hilbert inequality [14], the exponent $1+2d > 1$ suppresses mean-square off-diagonal sums on vertical windows $\Delta T \gg N(T)^{1-2d}$, recovering the classical Bohr–Landau [1] and Carlson [3] zero-density mechanism $N(\sigma, T) = o(T)$. By contrast, in Gallagher's Sobolev inequality [6], the unnormalized product $(\int_I |F|^2)^{1/2}(\int_I |F'|^2)^{1/2}$ grows at least as $\mathcal{O}(N(T)^{1-2d})$ for every choice of $\Delta T > 0$, reflecting Kronecker–Voronin phase alignment [21] and precluding pointwise positivity.

1.2. **Notation and conventions.**

For an integer $n \ge 1$, $\mu(n)$ denotes the Möbius function, $\Lambda(n)$ the classical von Mangoldt function, $\omega(n) := \sum_{p|n} 1$ the number of distinct prime factors of n, and $\mathrm{rad}(n) := \prod_{p|n} p$ its squarefree radical. Unsigned Stirling numbers of the first kind are written $\left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right]$. Vinogradov $(\ll)$ and Landau $(\mathcal{O}, \asymp)$ symbols carry absolute implied constants unless subscripts indicate dependence on parameters such as d or r.

2. Hyperbolic Diagonal Advantage, $\widetilde{\Lambda}_r(n)$ versus $\Lambda_r^{\mathrm{BS}}(n)$, and the $n > X$ Obstruction

Throughout this paper we write $L := 2\vartheta'(t) = \ln(t/2\pi) + \mathcal{O}(t^{-2})$ (or $L = \ln(t/2\pi)$ in asymptotic statements), $x_t := \mathrm{e}^{L/2} \sim \sqrt{t/2\pi}$, $N(t) := \lfloor\sqrt{t/2\pi}\rfloor$, $\omega_n(t) := \frac{1}{2}L - \ln n$, and $\phi_n(t) := t\ln n - \vartheta(t)$ for $1 \le n \le N(t)$, where $\vartheta(t) = \operatorname{Im}\ln\Gamma(\frac{1}{4} + \frac{\mathrm{i}t}{2}) - \frac{t}{2}\ln\pi$ is the Riemann–Siegel phase [19, 5]. Given real weights $\{b_n\}_{n=1}^{N(t)}$ with $b_1 = 1$, consider the symmetric harmonic model sum

$$S_N(d,t) := \sum_{n=1}^{N(t)} \frac{b_n}{\sqrt{n}}\Big(\mathrm{e}^{d\omega_n(t) - \mathrm{i}\phi_n(t)} + \mathrm{e}^{-d\omega_n(t) + \mathrm{i}\phi_n(t)}\Big), \tag{2.1}$$

whose squared modulus expands as

$$\frac{1}{2}\big|S_N(d,t)\big|^2 = \sum_{n=1}^{N(t)} \frac{b_n^2}{n}\big[\cosh(2d\omega_n) + \cos(2\phi_n)\big] + 2\sum_{1\le m<n\le N(t)} \frac{b_m b_n}{\sqrt{mn}}\Big[\cosh(2d\omega_{mn}^{+})\cos(\phi_n - \phi_m) + \cosh(2d\omega_{mn}^{-})\cos(\phi_m + \phi_n)\Big],$$

where

$$\omega_{mn}^{+}(t) := \frac{\omega_m(t) + \omega_n(t)}{2}, \qquad \omega_{mn}^{-} := \frac{\omega_m(t) - \omega_n(t)}{2} = \frac{1}{2}\ln\frac{n}{m}. \tag{2.2}$$

Applying $\frac{1}{2d}\partial_d$ to $\frac{1}{2}|S_N(d,t)|^2$ (holding $\omega_n(t)$ fixed in the model (2.1)) produces the diagonal sum $\mathcal{D}_N(d,t) = \frac{1}{d}\sum_{n=1}^{N(t)} \frac{b_n^2}{n}\omega_n(t)\sinh(2d\omega_n(t))$ and cross-weights $\mathcal{H}_{mn}^{\pm}(d,t) := \frac{1}{d}\omega_{mn}^{\pm}\sinh(2d\omega_{mn}^{\pm})$.

Proposition 2.1 (Termwise $(mn)^{-d} \le 2^{-d}$ Hyperbolic Damping)**.** *Let $d > 0$ and $t > 4\pi$ with $\omega_1(t) = \frac{1}{2}L > 0$. For every off-diagonal pair $1 \le m < n \le N(t)$ with $\omega_{mn}^{+}(t) > 0$,*

$$\sinh\big(2d\omega_{mn}^{+}(t)\big) = (mn)^{-d}\sinh\big(2d\omega_1(t)\big) - \sinh\big(d\ln(mn)\big)\mathrm{e}^{-dL}, \tag{2.3}$$

and consequently $\mathcal{H}_{mn}^{+}(d,t)/\mathcal{D}_{1,1}(d,t) < (1 - \frac{\ln(mn)}{L})(mn)^{-d} < (mn)^{-d} \le 2^{-d} < 1$, where $\mathcal{D}_{1,1}(d,t) := \frac{1}{d}\omega_1(t)\sinh(2d\omega_1(t))$.

Proof. Setting $A = 2d\omega_1(t) = dL > 0$ and $B = d\ln(mn) > 0$ ($mn \ge 2$), the identity $\sinh(A - B) = \mathrm{e}^{-B}\sinh(A) - \sinh(B)\mathrm{e}^{-A}$ yields (2.3). Since $\sinh(B)\mathrm{e}^{-A} > 0$ and $0 < \omega_{mn}^{+}(t)/\omega_1(t) = 1 - \frac{\ln(mn)}{L} < 1$, multiplying by $\omega_{mn}^{+}(t)/\omega_1(t)$ gives the stated inequality chain. □

Remark 2.2 (Termwise damping versus pair multiplicity). Proposition 2.1 bounds a *single* pair (m,n) against $(1,1)$. Summing $(mn)^{-1/2-d}$ over all $\binom{N(t)}{2}$ pairs still grows as $N(t)^{1-2d} \to +\infty$ for $0 < d < \frac{1}{2}$, just as in Paper V [11].

We now clarify the arithmetic properties of polynomial Möbius mollifiers [18, 10, 4] and resolve two subtleties: the distinction between $\widetilde{\Lambda}_r(n)$ and Bombieri–Selberg's $\Lambda_r^{\mathrm{BS}}(n)$, and the breakdown of $(\ln X)^{-r}$ scaling above the sieve horizon X.

Theorem 2.3 (Arithmetic Laws of $\widetilde{\Lambda}_r(n)$, Contrast with $\Lambda_r^{\mathrm{BS}}(n)$, and $n > X$ Breakdown)**.** *Let $r \in \mathbb{N}_{\ge 1}$ and $X > 1$. For $P_r(x) := 1 - (1-x)^r$, let $M_X^{(r)}(s) := \sum_{k\le X}\mu(k)P_r(\frac{\ln(X/k)}{\ln X})k^{-s}$ and write $\zeta(s)M_X^{(r)}(s) = \sum_{n=1}^{\infty} b_n^{(r)}(X)n^{-s}$ for $\sigma > 1$, so $b_n^{(r)}(X) = \sum_{k\mid n,\,k\le X}\mu(k)[1 - (\frac{\ln k}{\ln X})^r]$. Define the squarefree-divisor logarithmic sieve function and the Bombieri–Selberg generalized von Mangoldt function [2, 8] by*

$$\widetilde{\Lambda}_r(n) := -\sum_{k\mid n}\mu(k)(\ln k)^r, \qquad \Lambda_r^{\mathrm{BS}}(n) := \sum_{d\mid n}\mu(d)\Big(\ln\frac{n}{d}\Big)^r. \tag{2.4}$$

Then:

(i) ***Generating functions, Bombieri recursion, and algebraic divergence for*** $r \geq 2$***.*** *For* $\sigma > 1$, $\sum_{n=1}^{\infty} \widetilde{\Lambda}_r(n) n^{-s} = (-1)^{r+1} \zeta(s) \frac{\mathrm{d}^r}{\mathrm{d}s^r}(\frac{1}{\zeta(s)})$, *whereas* $\sum_{n=1}^{\infty} \Lambda_r^{\mathrm{BS}}(n) n^{-s} = (-1)^r \frac{\zeta^{(r)}(s)}{\zeta(s)}$. *For* $r = 1$, $\widetilde{\Lambda}_1(n) = \Lambda_1^{\mathrm{BS}}(n) = \Lambda(n)$. *For* $r \geq 1$, Λ_r^{BS} *satisfies* $\sum_{d|n} \Lambda_r^{\mathrm{BS}}(d) = (\ln n)^r$, *the recursion* $\Lambda_r^{\mathrm{BS}}(n) = \Lambda_{r-1}^{\mathrm{BS}}(n) \ln n + (\Lambda_{r-1}^{\mathrm{BS}} * \Lambda)(n)$, *and the sharp divisor bound* $0 \leq \Lambda_r^{\mathrm{BS}}(n) \leq (\ln n)^r$ *for all* $n \geq 1$. *For* $r = 2$, $\widetilde{\Lambda}_2(n) = \Lambda_2^{\mathrm{BS}}(n) - 2(\Lambda * \Lambda)(n)$*; in particular, at* $n = 8 = 2^3$, $\widetilde{\Lambda}_2(8) = (\ln 2)^2 = (\ln \mathrm{rad}(8))^2$ *whereas* $\Lambda_2^{\mathrm{BS}}(8) = 5(\ln 2)^2 > (\ln \mathrm{rad}(8))^2$.

(ii) ***Parity alternation, radical bound, and Stirling asymptotics of*** $\widetilde{\Lambda}_r(n)$***.*** *For* $n = p_1^{a_1} \cdots p_m^{a_m} \geq 2$ *with* $\omega(n) = m$ *and* $\mathrm{rad}(n) = p_1 \cdots p_m$, $\widetilde{\Lambda}_r(n) = 0$ *whenever* $m > r$, *while for* $1 \leq m \leq r$,

$$\begin{aligned} |\widetilde{\Lambda}_r(n)| = (-1)^{m-1} \widetilde{\Lambda}_r(n) &= \sum_{\substack{j_1 + \cdots + j_m = r \\ j_1, \ldots, j_m \geq 1}} \frac{\Gamma(r+1)}{\Gamma(j_1+1) \cdots \Gamma(j_m+1)} (\ln p_1)^{j_1} \cdots (\ln p_m)^{j_m} \\ &\leq \big(\ln \mathrm{rad}(n)\big)^r, \end{aligned} \tag{2.5}$$

and as $x \to \infty$, $\sum_{n \leq x, \, \omega(n) = m} |\widetilde{\Lambda}_r(n)| \sim \frac{1}{\Gamma(r)} \left[{r \atop m} \right] x (\ln x)^{r-1}$ *and* $\sum_{n \leq x} |\widetilde{\Lambda}_r(n)| \sim r \, x (\ln x)^{r-1}$, *where* $\left[{r \atop m} \right]$ *denotes the unsigned Stirling number of the first kind.*

(iii) ***Exact*** $(\ln X)^{-r}$ ***scaling on*** $2 \leq n \leq X$ ***versus*** $\mathcal{O}(1)$ ***breakdown for*** $n > X$***.*** *We have* $b_1^{(r)}(X) = 1$ *and*

$$b_n^{(r)}(X) = \frac{\widetilde{\Lambda}_r(n)}{(\ln X)^r} \quad (2 \leq n \leq X), \qquad b_p^{(r)}(X) = 1 \quad (\textit{every prime } p > X). \tag{2.6}$$

More generally, for $n > X$, $b_n^{(r)}(X) = \frac{\widetilde{\Lambda}_r(n)}{(\ln X)^r} - \sum_{k|n, \, k > X} \mu(k) [1 - (\frac{\ln k}{\ln X})^r]$ *fails to vanish even when* $\omega(n) > r$ *(for instance, at* $X = 10$, $r = 2$, $n = 30 = 2 \cdot 3 \cdot 5$, $\widetilde{\Lambda}_2(30) = 0$ *yet* $b_{30}^{(2)}(10) = -0.79869655\ldots \neq 0$*).*

Proof. (i) Differentiating $\zeta(s)^{-1} = \sum_{k=1}^{\infty} \mu(k) k^{-s}$ and $\zeta(s) = \sum_{m=1}^{\infty} m^{-s}$ r times on $\sigma > 1$ yields both Dirichlet generating functions. By Möbius inversion on $\Lambda_r^{\mathrm{BS}} = \mu * \ln^r$, $\sum_{d|n} \Lambda_r^{\mathrm{BS}}(d) = (\ln n)^r$. Writing $(\ln \frac{n}{d})^r = (\ln \frac{n}{d})^{r-1} (\ln n - \ln d)$ and using $-\mu(d) \ln d = (\mu * \Lambda)(d)$ gives $\Lambda_r^{\mathrm{BS}} = \Lambda_{r-1}^{\mathrm{BS}} \ln + \Lambda_{r-1}^{\mathrm{BS}} * \Lambda$; since $\Lambda_1^{\mathrm{BS}} = \Lambda \geq 0$, induction implies $\Lambda_r^{\mathrm{BS}}(n) \geq 0$, and the non-negative divisor sum $\sum_{d|n} \Lambda_r^{\mathrm{BS}}(d) = (\ln n)^r$ immediately forces $\Lambda_r^{\mathrm{BS}}(n) \leq (\ln n)^r$. Expanding $(\zeta^{-1})'' = -\zeta''/\zeta^2 + 2(\zeta')^2/\zeta^3$ gives $\widetilde{\Lambda}_2 = \Lambda_2^{\mathrm{BS}} - 2(\Lambda * \Lambda)$.

(ii) Since $\mu(k) = 0$ unless $k \mid \mathrm{rad}(n)$, inclusion–exclusion over subsets $S \subseteq \{1, \ldots, m\}$ in $(\sum_{i \in S} \ln p_i)^r$ cancels every multinomial tuple with any $j_i = 0$, leaving (2.5) for $1 \leq m \leq r$ and 0 for $m > r$. Comparing (2.5) with the full multinomial expansion of $(\ln p_1 + \cdots + \ln p_m)^r$ yields $|\widetilde{\Lambda}_r(n)| \leq (\ln \mathrm{rad}(n))^r$. By the Prime Number Theorem $\sum_{p^a \leq y} (\ln p)^j \sim y (\ln y)^{j-1}$ $(j \geq 1)$ and repeated Dirichlet convolution over m unordered prime-power factors, the leading coefficient is $\frac{r}{\Gamma(m+1)} \sum_{j_1 + \cdots + j_m = r, \, j_i \geq 1} (j_1 \cdots j_m)^{-1} = \frac{1}{\Gamma(r)} \left[{r \atop m} \right]$, whose sum over $1 \leq m \leq r$ is $\frac{\Gamma(r+1)}{\Gamma(r)} = r$.

(iii) For $2 \leq n \leq X$, every divisor $k \mid n$ satisfies $k \leq X$, so $\sum_{k|n} \mu(k) = 0$ yields $b_n^{(r)}(X) = -\sum_{k|n} \mu(k) (\frac{\ln k}{\ln X})^r = \widetilde{\Lambda}_r(n)/(\ln X)^r$. For a prime $p > X$, the only divisor $k \mid p$ with $k \leq X$ is $k = 1$, giving $b_p^{(r)}(X) = \mu(1)(1 - 0) = 1$. Subtracting the missing $k > X$ divisors from $\sum_{k|n} \mu(k) [1 - (\frac{\ln k}{\ln X})^r] = \widetilde{\Lambda}_r(n)/(\ln X)^r$ gives the exact discrepancy for general $n > X$. □

3. Exact Bilinear Rank-2 Factorization of Integrated Cross-Term Kernels

Within the symmetric harmonic model (2.1) (with $X \geq N(t)$ whenever $b_n = b_n^{(r)}(X)$ is represented as $\widetilde{\Lambda}_r(n)/(\ln X)^r$ on $2 \leq n \leq N(t)$), integrating the transverse hyperbolic modes over $u \in [0, d]$ yields a genuine rank-2 algebraic reduction.

Theorem 3.1 (Exact Bilinear Rank-2 Factorization and $\mathcal{O}(N)$ Reduction)**.** *Define the hyperbolic sinc* $\operatorname{sinch}(x) := \frac{\sinh x}{x}$ *(*$\operatorname{sinch}(0) := 1$*) and low-pass multiplier* $\mathcal{F}_{\mathrm{int}}(x) := \frac{\tanh(x/2)}{x/2}$ *(*$\mathcal{F}_{\mathrm{int}}(0) := 1$*). For any* $d > 0$ *and any frequencies* $\omega_m, \omega_n \in \mathbb{R}$ *with* $\Omega_{mn}^{\pm} := \omega_m \pm \omega_n = 2\omega_{mn}^{\pm}$*:*

(i) ***Single-mode low-pass identity.*** *For every* $\alpha > 0$*,*

$$\frac{2}{d^2}\int_0^d u\,\operatorname{sinch}(u\alpha)\,\mathrm{d}u = \frac{2(\cosh(d\alpha)-1)}{d^2\alpha^2} = \operatorname{sinch}^2\!\left(\frac{d\alpha}{2}\right) = \mathcal{F}_{\mathrm{int}}(d\alpha)\operatorname{sinch}(d\alpha) < \operatorname{sinch}(d\alpha). \tag{3.1}$$

(ii) ***Exact bilinear rank-2 factorization.*** *The integrated cross-term kernel factors identically as*

$$\boxed{\begin{aligned} w_{mn}^{\mathrm{int}}(d,t) &:= 2\big(\omega_{mn}^{+}(t)\big)^2\mathcal{F}_{\mathrm{int}}\big(d\Omega_{mn}^{+}(t)\big)\operatorname{sinch}\big(d\Omega_{mn}^{+}(t)\big) + 2\big(\omega_{mn}^{-}\big)^2\mathcal{F}_{\mathrm{int}}\big(d\Omega_{mn}^{-}\big)\operatorname{sinch}\big(d\Omega_{mn}^{-}\big) \\ &\equiv \frac{2\Big(\cosh\big(d\omega_m(t)\big)\cosh\big(d\omega_n(t)\big)-1\Big)}{d^2}. \end{aligned}} \tag{3.2}$$

(iii) $\mathcal{O}(N^2) \to \mathcal{O}(N)$ ***summation identity.*** *For any real or complex sequence* $\{a_n\}_{n=1}^N$*,*

$$\begin{aligned} &\sum_{1\leq m<n\leq N} a_m a_n\Big(\cosh(d\omega_m)\cosh(d\omega_n)-1\Big) \\ &\qquad \equiv \frac{1}{2}\left[\left(\sum_{n=1}^N a_n\cosh(d\omega_n)\right)^2 - \left(\sum_{n=1}^N a_n\right)^2 - \sum_{n=1}^N a_n^2\sinh^2(d\omega_n)\right]. \end{aligned} \tag{3.3}$$

Proof. (i) Since $2u\operatorname{sinch}(u\alpha) = \frac{2}{\alpha}\sinh(u\alpha)$, integrating over $u \in [0,d]$ and using $\cosh(d\alpha)-1 = 2\sinh^2(d\alpha/2)$ and $\sinh(d\alpha) = 2\sinh(d\alpha/2)\cosh(d\alpha/2)$ yields (3.1), with $\tanh y < y$ for $y > 0$.

(ii) By (3.1), $2\omega^2\mathcal{F}_{\mathrm{int}}(2d\omega)\operatorname{sinch}(2d\omega) = \frac{\cosh(2d\omega)-1}{d^2}$. Applying this to $\omega = \omega_{mn}^{\pm} = \frac{1}{2}(\omega_m \pm \omega_n)$ and using $\cosh(d\omega_m + d\omega_n) + \cosh(d\omega_m - d\omega_n) = 2\cosh(d\omega_m)\cosh(d\omega_n)$ yields (3.2).

(iii) Expanding $2\sum_{1\leq m<n\leq N} x_m x_n = (\sum_{n=1}^N x_n)^2 - \sum_{n=1}^N x_n^2$ for $x_n = a_n\cosh(d\omega_n)$ and $x_n = a_n$, subtracting, and using $\cosh^2(d\omega_n) - 1 = \sinh^2(d\omega_n)$ gives (3.3). □

4. Euclidean Representation and the $\sinh(dL/2)$ Reverse-Triangle Obstruction

We now show how the -1 offset in (3.2) cancels against the $d = 0$ boundary term of (2.1), and prove why the resulting Euclidean reverse-triangle inequality *cannot* certify positivity of $|S_N(d,t)|^2$ on $0 < d < \frac{1}{2}$.

Theorem 4.1 (Euclidean Representation and Reverse-Triangle Collapse)**.** *For* $1 \leq n \leq N(t)$*, define the Euclidean harmonic vector* $\psi_n(d,t) \in \mathbb{R}^2$ *by*

$$\psi_n(d,t) := \frac{b_n}{\sqrt{n}}\Big(\cosh\big(d\omega_n(t)\big)\cos\phi_n(t),\ -\sinh\big(d\omega_n(t)\big)\sin\phi_n(t)\Big), \qquad \Psi_{\geq 2}(d,t) := \sum_{n=2}^{N(t)}\psi_n(d,t), \tag{4.1}$$

with $b_1 = 1$*. Then for every* $d > 0$*:*

(i) ***Exact Euclidean norm and -1 boundary cancellation.*** *In (2.1), $\frac{1}{2}S_N(d,t) = \sum_{n=1}^{N(t)}(\psi_{n,1} + \mathrm{i}\psi_{n,2})$, so*

$$\frac{1}{4}\big|S_N(d,t)\big|^2 \equiv \big\|\psi_1(d,t) + \Psi_{\geq 2}(d,t)\big\|^2 \geq \Big(\max\big(0,\ \|\psi_1(d,t)\| - \|\Psi_{\geq 2}(d,t)\|\big)\Big)^2, \tag{4.2}$$

and for every pair (m,n), $\langle \psi_m(d,t), \psi_n(d,t)\rangle - \langle \psi_m(0,t), \psi_n(0,t)\rangle = \frac{b_m b_n}{2\sqrt{mn}}[(\cosh(d\Omega_{mn}^+) - 1)\cos(\phi_n - \phi_m) + (\cosh(d\Omega_{mn}^-) - 1)\cos(\phi_m + \phi_n)]$.

(ii) ***True minimum $\|\psi_1(d,t)\|_{\min} = \sinh(dL/2)$ versus $\cosh(dL/2)$.*** *Because $\omega_1(t) = \frac{1}{2}L$ and $b_1 = 1$,*

$$\big\|\psi_1(d,t)\big\|^2 = \cosh^2\!\left(\frac{dL}{2}\right)\cos^2\phi_1(t) + \sinh^2\!\left(\frac{dL}{2}\right)\sin^2\phi_1(t) = \sinh^2\!\left(\frac{dL}{2}\right) + \cos^2\phi_1(t). \tag{4.3}$$

Hence as t varies, $\|\psi_1(d,t)\|$ oscillates between its maximum $\cosh(dL/2)$ (at $\cos\phi_1(t) = \pm 1$) and its minimum $\|\psi_1(d,t)\|_{\min} = \sinh(dL/2)$ (attained whenever $\vartheta(t) \equiv \frac{\pi}{2} \pmod{\pi}$, i.e. $\sim \frac{T}{2\pi}\ln T$ times on $[0,T]$). Substituting $\cosh(dL/2)$ into a lower bound or omitting $\max(0,\cdot)$ in (4.2) before squaring is invalid: $(1-Y)^2 > 0$ holds algebraically even when $Y > 1$ and $\psi_1 + \Psi_{\geq 2} = 0$.

(iii) ***Asymptotic collapse of the phase-blind lower bound on $0 < d < \frac{1}{2}$.*** *Under $b_n = b_n^{(r)}(X) = \widetilde{\Lambda}_r(n)/(\ln X)^r$ with $X = N(t)$ ($r \geq 1$), the triangle-inequality envelope $\Sigma_{\cosh}^{(r)}(d,t) := \sum_{n=2}^{N(t)} \frac{|b_n|}{\sqrt{n}}\cosh(d\omega_n(t)) \geq \|\Psi_{\geq 2}(d,t)\|$ satisfies, for every fixed $d \in (0,\frac{1}{2})$ as $t \to \infty$,*

$$\frac{\Sigma_{\cosh}^{(r)}(d,t)}{\sinh(dL/2)} \sim \frac{4r}{1-4d^2}\,\frac{N(t)^{\frac{1}{2}-d}}{\ln N(t)} \longrightarrow +\infty, \tag{4.4}$$

so the phase-blind lower bound $\max(0, \sinh(dL/2) - \Sigma_{\cosh}^{(r)}(d,t))$ equals 0 identically for all sufficiently large t.

Proof. (i) Writing $\mathrm{e}^{\pm(d\omega_n - \mathrm{i}\phi_n)} = (\cosh d\omega_n \pm \sinh d\omega_n)(\cos\phi_n \mp \mathrm{i}\sin\phi_n)$ and summing the two conjugate branches gives $\frac{1}{2}S_N(d,t) = \sum_{n=1}^{N(t)}\frac{b_n}{\sqrt{n}}(\cosh(d\omega_n)\cos\phi_n - \mathrm{i}\sinh(d\omega_n)\sin\phi_n)$, whose squared modulus is $\|\psi_1 + \Psi_{\geq 2}\|^2$. Since $\|u+v\| \geq |\|u\| - \|v\||$ can vanish whenever $\|v\| \geq \|u\|$, a phase-blind upper bound $\|\Psi_{\geq 2}\| \leq B$ implies only $\|\psi_1 + \Psi_{\geq 2}\| \geq \max(0, \|\psi_1\| - B)$, proving (4.2).

(ii) Using $\cosh^2(dL/2) = 1 + \sinh^2(dL/2)$ and $\cos^2\phi_1 + \sin^2\phi_1 = 1$ yields (4.3). Since $\vartheta(t) \sim \frac{t}{2}\ln\frac{t}{2\pi\mathrm{e}}$ is continuous and strictly increasing for $t \geq 7$, $\cos\phi_1(t) = \cos\vartheta(t) = 0$ occurs at every half-integer multiple of π.

(iii) By Theorem 2.3(ii), $\sum_{n\leq x}|\widetilde{\Lambda}_r(n)| \sim rx(\ln x)^{r-1}$. Writing $\cosh(d\omega_n(t)) = \frac{1}{2}(x_t^d n^{-d} + x_t^{-d}n^d)$ and integrating by parts on $[2, N(t)]$ for fixed $d \in (0,\frac{1}{2})$ gives

$$\begin{aligned}\sum_{n=2}^{N(t)}\frac{|\widetilde{\Lambda}_r(n)|}{\sqrt{n}}\cosh(d\omega_n(t)) &\sim \frac{r}{2}\left(\frac{1}{\frac{1}{2}-d} + \frac{1}{\frac{1}{2}+d}\right)N(t)^{1/2}(\ln N(t))^{r-1} \\ &= \frac{2r}{1-4d^2}\,N(t)^{1/2}(\ln N(t))^{r-1}.\end{aligned}$$

Dividing by $(\ln X)^r = (\ln N(t))^r$ and by $\sinh(dL/2) \sim \frac{1}{2}x_t^d \sim \frac{1}{2}N(t)^d$ yields (4.4), which diverges to $+\infty$ because $\frac{1}{2} - d > 0$. □

TABLE 1. Exact values of $\|\psi_1\|_{\min} = \sinh(dL/2)$ (versus $\|\psi_1\|_{\max} = \cosh(dL/2)$), tail envelope $\Sigma^{(2)}_{\cosh}(d,t)$, and ratio $\Sigma^{(2)}_{\cosh}/\sinh(dL/2) > 1$ at $d = 0.20$, $r = 2$, $X = N(t)$, showing that the true reverse-triangle lower bound vanishes identically.

Ordinate t	$N(t)$	$\|\psi_1\|_{\min}$	$\|\psi_1\|_{\max}$	$\Sigma^{(2)}_{\cosh}(d,t)$	Ratio $\Sigma^{(2)}_{\cosh}/\|\psi_1\|_{\min}$	True LB
100	3	0.2803	1.0385	0.8625	3.0772	0.0000
1,000	12	0.5290	1.1313	1.3017	2.4607	0.0000
10,000	39	0.8059	1.2843	2.5271	3.1359	0.0000
10^6	398	1.5054	1.8073	7.8424	5.2094	0.0000

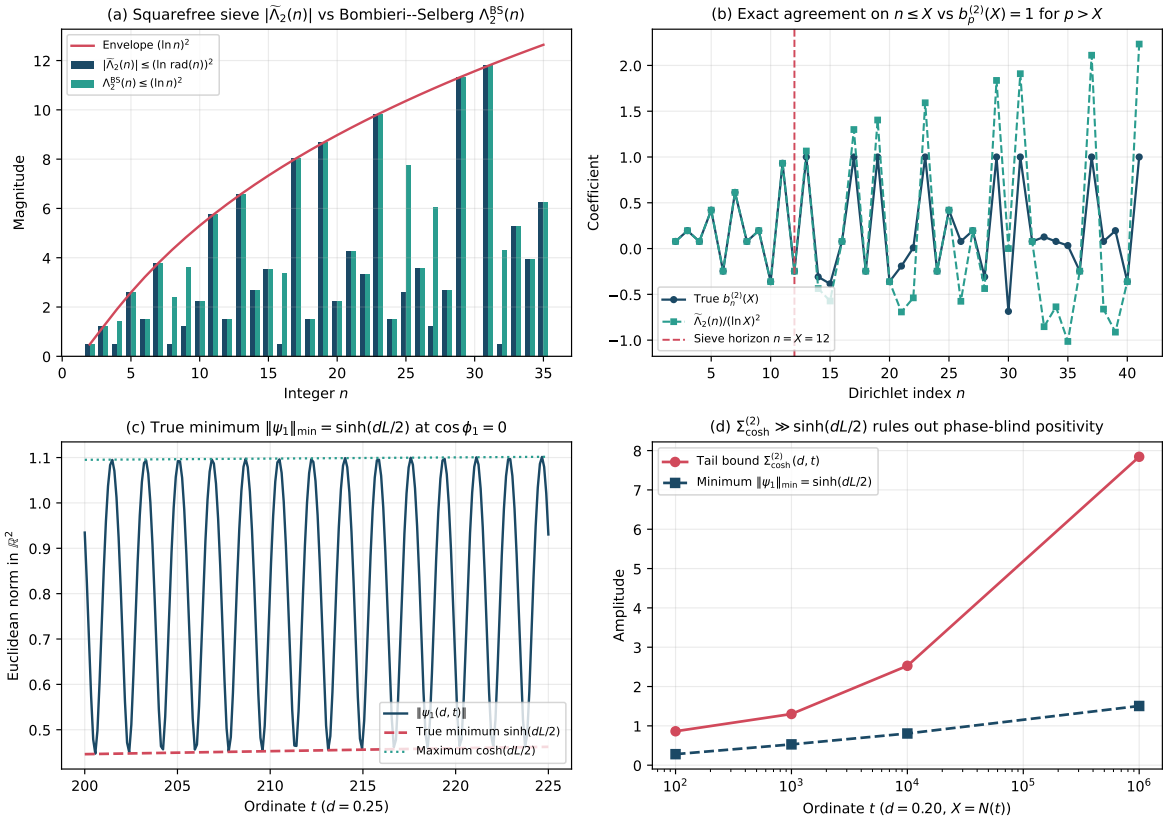


FIGURE 1. **(a)** Comparison of the squarefree-divisor sieve magnitude $|\widetilde{\Lambda}_2(n)| \le (\ln \mathrm{rad}(n))^2$ against Bombieri–Selberg's $\Lambda_2^{\mathrm{BS}}(n) \le (\ln n)^2$. **(b)** Exact identity $b_n^{(2)}(X) = \widetilde{\Lambda}_2(n)/(\ln X)^2$ for $2 \le n \le X = 12$ versus $\mathcal{O}(1)$ breakdown $b_p^{(2)}(X) = 1$ at primes $p > X$. **(c)** Oscillation of $\|\psi_1(d,t)\|$ down to $\|\psi_1\|_{\min} = \sinh(dL/2)$ at $\cos\phi_1(t) = 0$. **(d)** Divergence of $\Sigma^{(2)}_{\cosh}(d,t)$ past $\sinh(dL/2)$ ($d = 0.20$, $X = N(t)$).

5. PREISSMANN–MONTGOMERY–VAUGHAN L^2 WINDOW BOUNDS VERSUS POINTWISE GALLAGHER EXPLOSION

Passing from phase-blind L^1 sums to vertical L^2 averages shifts the forward Dirichlet exponent from $\frac{1}{2} + d < 1$ to $2(\frac{1}{2} + d) = 1 + 2d > 1$. We now state the mean-square inequality with

Preissmann's true sharp constant [16] and prove why Gallagher's Sobolev inequality [6] cannot convert this L^2 damping into a pointwise bound.

Theorem 5.1 (Preissmann–Montgomery–Vaughan L^2 Bounds and Failure of Pointwise Gallagher Transfer)**.** *Let $F(\tau) := \sum_{n=2}^{N} a_n n^{-\mathrm{i}\tau}$ with $a_n = b_n^{(r)}(X) n^{-(1/2+d)}$, $d \in (0, \frac{1}{2})$, $N = N(T) = \lfloor \sqrt{T/2\pi} \rfloor$, and $X = N(T)$. Define $\mathcal{E}_{\mathrm{diag}}(d,T) := \sum_{n=2}^{N} |a_n|^2$, $\mathcal{E}_{\mathrm{deriv}}(d,T) := \sum_{n=2}^{N} (\ln n)^2 |a_n|^2$, and $\mathcal{W}_{\mathrm{MV}}(d,T) := \sum_{n=2}^{N} (n+\frac{1}{2}) |a_n|^2$. Then:*

(i) ***True Preissmann constant and mean-square window suppression.*** *By the Montgomery–Vaughan Hilbert inequality [14] with Preissmann's sharp constant [16]*

$$c_{\mathrm{MV}} := \pi \sqrt{1 + \frac{2}{3}\sqrt{\frac{6}{5}}} = 4.13247436208154\ldots, \tag{5.1}$$

and spacing $\delta_n = \ln(n+1) - \ln n > (n+\frac{1}{2})^{-1}$, every interval $I = [t_0 - \Delta T/2, t_0 + \Delta T/2]$ of width $\Delta T > 0$ satisfies

$$\frac{1}{\Delta T} \int_I |F(\tau)|^2 \, \mathrm{d}\tau \le \mathcal{E}_{\mathrm{diag}}(d,T) + \frac{2c_{\mathrm{MV}}}{\Delta T} \, \mathcal{W}_{\mathrm{MV}}(d,T). \tag{5.2}$$

Because $1 + 2d > 1$ while $\sum_{n \le x} \widetilde{\Lambda}_r(n)^2 \asymp_r x (\ln x)^{2r-1}$, we obtain $\mathcal{E}_{\mathrm{diag}}(d,T) = \mathcal{O}_{d,r}((\ln N)^{-2r}) \to 0$ and $\mathcal{W}_{\mathrm{MV}}(d,T) \asymp_{d,r} N^{1-2d} (\ln N)^{-1}$ as $T \to \infty$. Consequently, for every width $\Delta T \ge \Delta T_{\mathrm{MV}}(d,T) := 2c_{\mathrm{MV}} \mathcal{W}_{\mathrm{MV}} / \mathcal{E}_{\mathrm{diag}} \asymp_{d,r} N(T)^{1-2d} (\ln N)^{2r-1}$, the mean square obeys $\frac{1}{\Delta T} \int_I |F(\tau)|^2 \, \mathrm{d}\tau \le 2\mathcal{E}_{\mathrm{diag}}(d,T) \to 0$.

(ii) ***Divergence of the Gallagher Sobolev bound for all $\Delta T > 0$.*** *Applying Gallagher's inequality [6]*

$$|F(t_0)|^2 \le \frac{1}{\Delta T} \int_I |F(\tau)|^2 \, \mathrm{d}\tau + \left(\int_I |F(\tau)|^2 \, \mathrm{d}\tau \right)^{1/2} \left(\int_I |F'(\tau)|^2 \, \mathrm{d}\tau \right)^{1/2} \tag{5.3}$$

together with the Preissmann–Montgomery–Vaughan bounds $I_0(\Delta T) := \Delta T \, \mathcal{E}_{\mathrm{diag}} + 2c_{\mathrm{MV}} \mathcal{W}_{\mathrm{MV}}$ on $\int_I |F|^2$ and $I_1(\Delta T) := \Delta T \, \mathcal{E}_{\mathrm{deriv}} + 2c_{\mathrm{MV}} \mathcal{W}'_{\mathrm{MV}}$ on $\int_I |F'|^2$ ($\mathcal{W}'_{\mathrm{MV}} := \sum_{n=2}^{N} (n+\frac{1}{2}) (\ln n)^2 |a_n|^2 \asymp_{d,r} N^{1-2d} \ln N$) yields, at $\Delta T = \Delta T_{\mathrm{MV}}$, the divergent Sobolev cross-term $\sqrt{I_0(\Delta T_{\mathrm{MV}}) I_1(\Delta T_{\mathrm{MV}})} \ge \Delta T_{\mathrm{MV}} \sqrt{\mathcal{E}_{\mathrm{diag}} \mathcal{E}_{\mathrm{deriv}}} \asymp_{d,r} N(T)^{1-2d} \to +\infty$. More generally, minimizing over all *$\Delta T > 0$ gives*

$$\inf_{\Delta T > 0} \left[\frac{I_0(\Delta T)}{\Delta T} + \sqrt{I_0(\Delta T) I_1(\Delta T)} \right] > 2c_{\mathrm{MV}} \sqrt{\mathcal{W}_{\mathrm{MV}}(d,T) \, \mathcal{W}'_{\mathrm{MV}}(d,T)} \asymp_{d,r} N(T)^{1-2d} \longrightarrow +\infty. \tag{5.4}$$

Proof. (i) Integrating $|F(\tau)|^2 = \sum_{n=2}^{N} |a_n|^2 + \sum_{2 \le m \ne n \le N} a_m \bar{a}_n \mathrm{e}^{-\mathrm{i}\tau(\ln m - \ln n)}$ over I expresses the off-diagonal integral as the difference of two generalized Hilbert forms $\frac{1}{\mathrm{i}} \sum_{m \ne n} \frac{u_m^{\mp} \overline{u_n^{\mp}}}{\ln m - \ln n}$ with $|u_n^{\pm}| = |a_n|$. Preissmann [16, Theorem 1] proved that for any distinct real frequencies λ_n with $\delta_n = \min_{m \ne n} |\lambda_m - \lambda_n|$, $\left| \sum_{m \ne n} \frac{u_m \bar{u}_n}{\lambda_m - \lambda_n} \right| \le c_{\mathrm{MV}} \sum_n \delta_n^{-1} |u_n|^2$ with c_{MV} given by (5.1). Since $\ln(n+1) - \ln n = \ln(\frac{n+1/2+1/2}{n+1/2-1/2}) > (n+\frac{1}{2})^{-1}$, combining both endpoints gives (5.2). Because $\sum_{n=2}^{\infty} \widetilde{\Lambda}_r(n)^2 n^{-(1+2d)} < \infty$ for $d > 0$, dividing by $(\ln N)^{2r}$ gives $\mathcal{E}_{\mathrm{diag}} \asymp_{d,r} (\ln N)^{-2r}$, while Stieltjes summation of $n^{1-(1+2d)} = n^{-2d}$ against $\widetilde{\Lambda}_r(n)^2$ gives $\mathcal{W}_{\mathrm{MV}} \asymp_{d,r} N^{1-2d} (\ln N)^{2r-1} (\ln N)^{-2r} = N^{1-2d} (\ln N)^{-1}$.

(ii) Integrating $\int_I \psi'(\tau) |F(\tau)|^2 \, \mathrm{d}\tau$ by parts with sawtooth $\psi(\tau) = \frac{\tau - t_0}{\Delta T} + \frac{1}{2} \mathrm{sgn}(t_0 - \tau)$ ($|\psi| \le \frac{1}{2}$) yields (5.3). For every $\Delta T > 0$, $I_0(\Delta T) > 2c_{\mathrm{MV}} \mathcal{W}_{\mathrm{MV}}$ and $I_1(\Delta T) > 2c_{\mathrm{MV}} \mathcal{W}'_{\mathrm{MV}}$, so $\sqrt{I_0(\Delta T) I_1(\Delta T)} > 2c_{\mathrm{MV}} \sqrt{\mathcal{W}_{\mathrm{MV}} \mathcal{W}'_{\mathrm{MV}}} \asymp_{d,r} N(T)^{1-2d} (\ln N)^{-1/2} (\ln N)^{1/2} = N(T)^{1-2d}$, establishing (5.4). □

TABLE 2. Mean-square window damping versus Gallagher Sobolev explosion at $d = 0.20$, $r = 2$, $X = N(T)$ using Preissmann's true constant $c_{\mathrm{MV}} = \pi\sqrt{1+\frac{2}{3}\sqrt{\frac{6}{5}}} \approx 4.132474$.

Height T	$N(T)$	$\mathcal{E}_{\mathrm{diag}}(d,T)$	Critical window ΔT_{MV}	Window mean $\le 2\mathcal{E}_{\mathrm{diag}}$	Gallagher Sobolev $\sqrt{I_0 I_1}$
100	3	0.27484	27.12	0.54969	15.401
1,000	12	0.09885	67.07	0.19770	27.346
10,000	39	0.07867	168.76	0.15734	80.455
10^6	398	0.04528	1,109.22	0.09056	495.548

Remark 5.2 (Bohr–Landau / Carlson zero density versus Kronecker–Voronin phase alignment). Theorem 5.1 cleanly separates what $1+2d>1$ proves from what it cannot prove:

(a) *Mean-square zero-density consequence.* On $[T,2T]$ (where $\Delta T = T \gg N(T)^{1-2d}$, or up to TX^2 with a genuine mollifier $M_X(s)$ for $X = T^\theta$, $\theta < \frac{1}{2}$), $1+2d>1$ together with Montgomery–Vaughan controls the mean square of $1-\zeta(s)M_X(s)$ and, via Littlewood's lemma, yields $N(\sigma,T) = o(T)$ (and Carlson's $N(\sigma,T) \ll T^{4\sigma(1-\sigma)+\varepsilon}$ [1, 3, 20, 7]).

(b) *Pointwise obstruction.* The failure of (5.4) to beat $\mathcal{O}(N(T)^{1-2d})$ is not an artifact of Gallagher's inequality: because $\{\ln p\}$ is $\mathbb{Q}$-linearly independent, Kronecker's theorem (and Voronin's universality theorem [21]) implies that for any fixed finite truncation and phases, rare heights t_0 exist where $n^{-\mathrm{i}t_0}$ aligns coherently across primes, realizing amplitudes far larger than the L^2 root-mean-square.

6. Scope, Boundary Conditions, and Open Questions

We summarize the exact boundary between the unconditional algebraic identities proved above and the analytic obstructions in the critical strip:

(1) *Symmetric harmonic model versus mollified approximate functional equations.* Theorems 3.1 and 4.1(i) hold identically for any finite symmetric harmonic sum (2.1). When $\zeta(s)$ is multiplied by a Dirichlet polynomial mollifier $M_X^{(r)}(s)$ of length X, Theorem 2.3(iii) shows that $b_n^{(r)}(X) = \widetilde{\Lambda}_r(n)/(\ln X)^r$ requires $n \le X$, while Paper VI [12] proves that $\zeta(s)M_X^{(r)}(s)$ breaks the functional equation of $\Xi(z)$ and introduces an incomplete convolution tail up to $N(t)X$ together with an unsieved reflected branch.

(2) *Zero-density window control versus pointwise zero-free regions.* Theorem 4.1(ii)–(iii) rules out phase-blind reverse-triangle positivity on $0<d<\frac{1}{2}$, and Theorem 5.1 proves that Preissmann–Montgomery–Vaughan L^2 bounds yield Bohr–Landau / Carlson mean-square suppression on windows $\Delta T \gg N(T)^{1-2d}$ while Gallagher's Sobolev transfer diverges at scale $N(T)^{1-2d}$ for all $\Delta T > 0$. Bridging the gap between vertical L^2 zero-density estimates and pointwise bounds in $0<d<\frac{1}{2}$ remains equivalent to controlling coherent Dirichlet-polynomial phase alignment beyond the Riemann–Siegel horizon.

Appendix A. Computational Verification Protocol

All algebraic identities, counterexamples, interval enclosures, and figures in this paper are certified by the standalone Python script `scripts/verify_paper8_bilinear_rank2_wronskian.py` (`mpmath.mp.dps = 50`, `mpmath.iv.dps = 35`):

(1) *Canonical Wronskian and sieve identities.* Certifies $|W(d,t) - 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y| = 1.54\times10^{-33}$ at $(d,t) = (0.25, 3.5)$, Bombieri's recursion residual 3.42×10^{-49} ($r = 2, 3,$

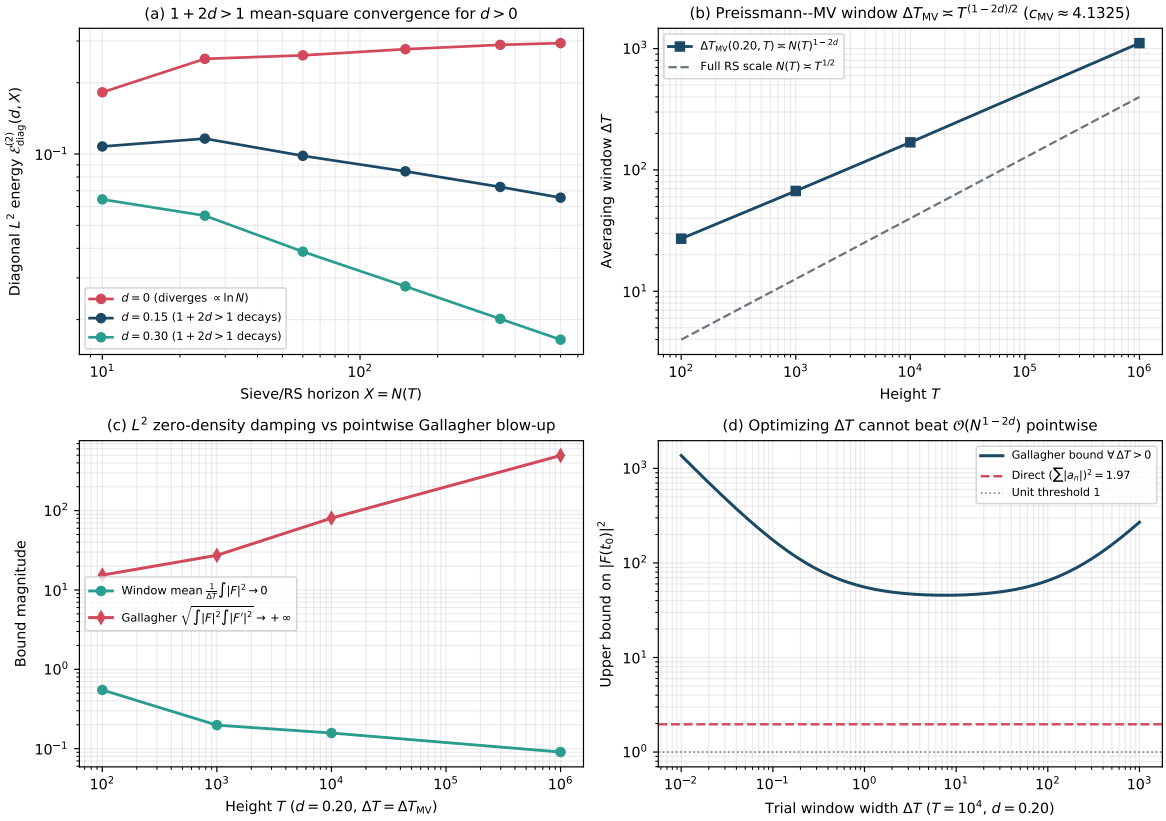


FIGURE 2. **(a)** Convergence and $(\ln X)^{-4}$ decay of $\mathcal{E}^{(2)}_{\mathrm{diag}}(d, X)$ for $d > 0$ via $1 + 2d > 1$, versus logarithmic growth at $d = 0$. **(b)** Scaling of the Preissmann–Montgomery–Vaughan window $\Delta T_{\mathrm{MV}}(0.20, T) \asymp N(T)^{1-2d}$ ($c_{\mathrm{MV}} \approx 4.1325$). **(c)** Simultaneous decay of the L^2 window mean $\frac{1}{\Delta T}\int_I |F|^2 \to 0$ and divergence of Gallagher's Sobolev term $\sqrt{\int_I |F|^2 \int_I |F'|^2} \to +\infty$. **(d)** Sweep over all $\Delta T > 0$ at $T = 10^4$, $d = 0.20$, confirming (5.4).

$n \le 80$), $|\widetilde{\Lambda}_2 - (\Lambda_2^{\mathrm{BS}} - 2(\Lambda * \Lambda))| \le 1.18 \times 10^{-49}$, exact agreement $b_n^{(2)}(10) = \widetilde{\Lambda}_2(n)/(\ln 10)^2$ on $2 \le n \le 10$ (1.34×10^{-51}), and $n > X$ breakdown $b_{11}^{(2)}(10) = 1$, $b_{30}^{(2)}(10) = -0.79869655$.

(2) *Rank-2 reduction and reverse-triangle obstruction.* Certifies (3.3) (1.13×10^{-51}), (4.2) (2.67×10^{-51}), and Table 1.

(3) *Preissmann constant and Gallagher divergence.* Encloses Preissmann's constant $c_{\mathrm{MV}} = \pi\sqrt{1 + \frac{2}{3}\sqrt{\frac{6}{5}}} \in [4.132474362081, 4.132474362082]$ in `mpmath.iv` and reproduces Table 2.

USE OF AI AND COMPUTATIONAL VERIFICATION

An AI assistant (**Gemini**, Google DeepMind) was used to audit earlier drafts, identify the reverse-triangle cosh-versus-sinh error, the $n > X$ sieve-horizon breakdown, the Preissmann constant and journal citation error, and the Gallagher Sobolev divergence, and to maintain the verification script `scripts/verify_paper8_bilinear_rank2_wronskian.py`. All mathematical proofs, inequalities, and numerical assertions were verified analytically and computationally (exit code 0).

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Independent Researcher, Belgium

Email address: thijs.missiaen@gmail.com